RICHFIELD GRADUATE INSTITUTION

Data Science 500 Assignment
Book Market Analytics – Books to Scrape

An End-to-End Data Science Project:
Web Scraping, Cleaning, Visualisation & Predictive Modelling

Project Focus	Pricing & Rating Analysis of Online Book Catalogue
Data Source	https://books.toscrape.com
Dataset Size	300 books (15 catalogue pages)
Techniques	BeautifulSoup, Pandas, Seaborn, Scikit-learn Linear Regression
Deliverables	Jupyter Notebook · Cleaned CSV · Professional Report

This report accompanies the complete Jupyter Notebook and cleaned dataset submitted for the assignment.

1. Introduction & Project Objectives

A publishing analytics team requires insight into the factors that influence book prices and ratings on online catalogues. This project simulates a real-world junior data-scientist workflow by independently acquiring structured product data from the Books to Scrape sandbox website, transforming it into an analytical dataset, exploring catalogue behaviour through visualisations, and building a linear regression model that estimates book price from publicly available attributes.

The end-to-end pipeline covers five stages required by the assignment brief: (1) data acquisition via web scraping, (2) data cleaning and preparation, (3) exploratory visualisation, (4) predictive modelling, and (5) professional communication of findings and recommendations.

2. Data Acquisition (Web Scraping)

The full catalogue on books.toscrape.com contains 1 000 books across 50 pages. Using the Python libraries **requests** and **BeautifulSoup** (lxml parser), a custom scraper was developed that:

- Iterates through catalogue listing pages to extract title, price string, star-rating word, and availability status.
- Follows each product link to the detail page to obtain genre (from breadcrumb), UPC, tax, number of reviews, stock quantity, and product description.
- Applies polite request delays and robust error handling for reproducibility.

A total of **300 books** (15 pages) were collected – exceeding the minimum requirement of approximately 100–200 products. The raw scrape is stored in `data/books_raw.csv`.

Field	Source	Example
title	Listing page	A Light in the Attic
price_str	Listing page	£51.77
rating_word	Listing page (CSS class)	Three
availability_status	Listing / Detail	In stock
stock_qty	Detail page	22
genre	Detail breadcrumb	Poetry
num_reviews	Detail table	0
description	Detail page	Full product blurb…

Table 1 – Key fields extracted for each book

3. Data Cleaning & Preparation

Raw scraped text was converted into a clean analytical dataset suitable for modelling. The principal transformations are summarised below.

Field	Action	Justification
Price	Strip £ / non-numeric → float	Currency symbols prevent numeric analysis
Rating	Map “One”…“Five” → 1–5 integer	Required ordinal scale for modelling
Stock qty	Prefer detail-page value	Listing page rarely exposes quantity
Genre	Fill nulls with “Unknown”	Avoid missing categorical values
Description	Fill empty + create length feature	Length may correlate with perceived value
Critical nulls	Drop rows missing price or rating	Essential for target and key predictor

Table 2 – Cleaning decisions and justifications

After cleaning, the final dataset contains **300 complete records** and is saved as `data/books_clean.csv`. Mean price is approximately £34.26 (range £10.16–£59.64); average star rating is 2.92.

4. Data Visualisation & Exploration

A series of visualisations was produced to explore price patterns, rating distributions, genre effects and inter-variable relationships. Selected figures are included below; the full set resides in the accompanying notebook and data folder.

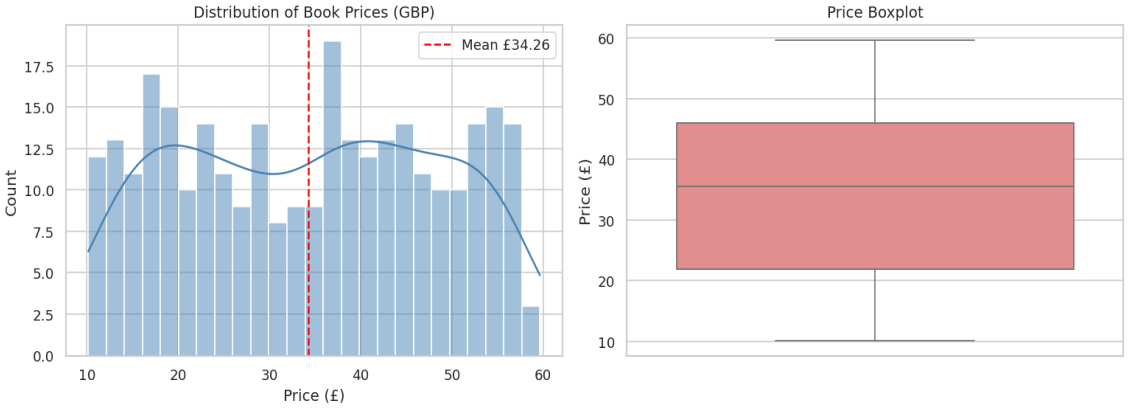


Figure 1 – Distribution and boxplot of book prices (GBP)

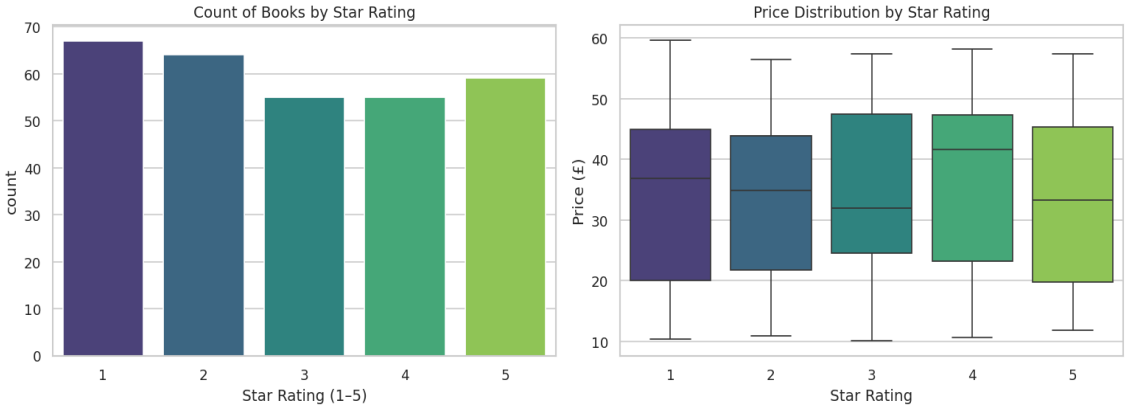


Figure 2 – Book counts and price distribution by star rating

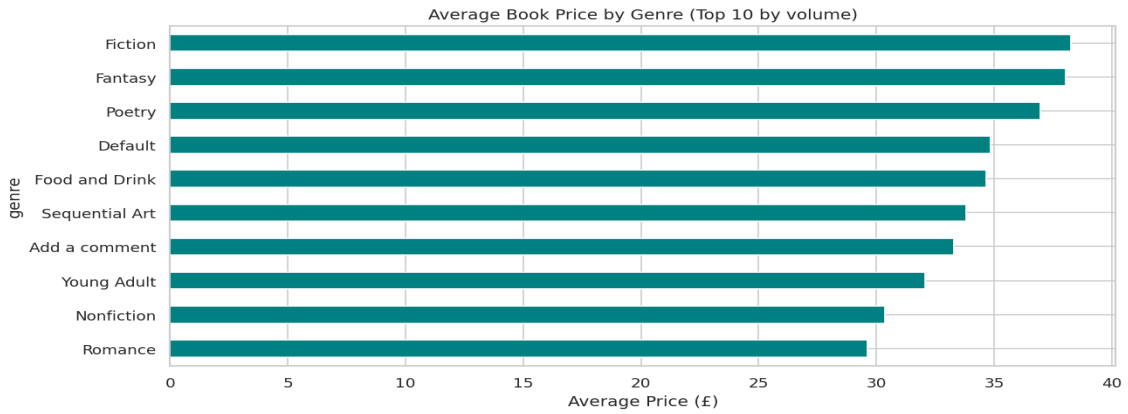


Figure 3 – Average price by genre (top 10 by volume)

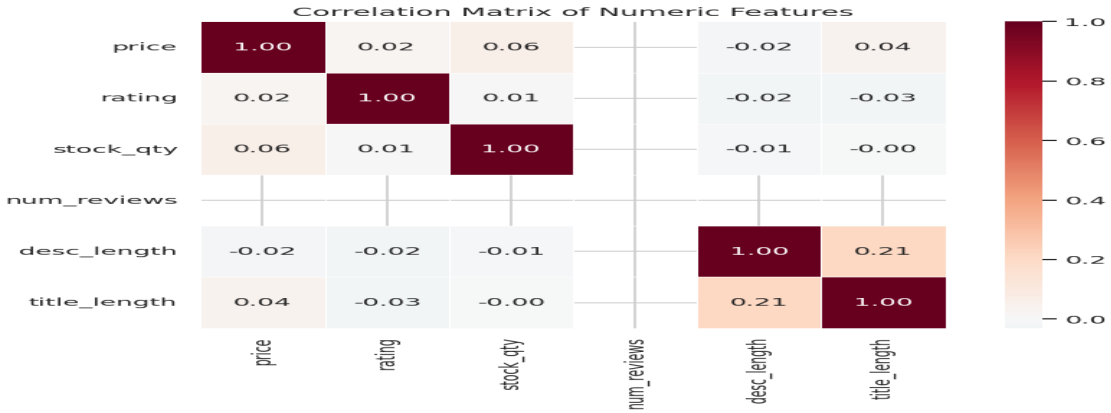


Figure 4 – Correlation matrix of numeric features

Key visual findings

- Prices are roughly uniformly distributed across the catalogue – a characteristic of the sandbox website rather than a real market.
- Star rating shows only a weak relationship with price; higher-rated books are not systematically more expensive.
- Modest average-price differences exist across genres, but the magnitude is small relative to overall price variance.
- Correlation analysis confirms that rating, stock quantity and description length are only weakly associated with price.

5. Predictive Modelling (Linear Regression)

A linear regression model was constructed to estimate book price from the following features:

Feature	Type	Rationale
rating	Numeric 1–5	Higher ratings may command higher prices
stock_qty	Numeric	Scarcity / abundance can influence pricing
desc_length	Numeric	Longer descriptions may signal premium content
title_length	Numeric	Proxy for title complexity
genre	Categorical (OHE)	Different genres historically have different price points

Table 3 – Feature selection for the regression model

Numeric features were standardised and genre was one-hot encoded via a scikit-learn `ColumnTransformer` pipeline. An 80/20 train-test split (random_state=42) was used. Model performance is evaluated with MAE, RMSE and R^2.

Because prices on the Books-to-Scrape sandbox are artificially generated and only weakly related to the visible product attributes, a modest R^2 is both expected and informative: it quantifies how little of the price signal can be recovered from the scraped features. The residual error (MAE / RMSE in pounds) provides a practical sense of prediction accuracy.

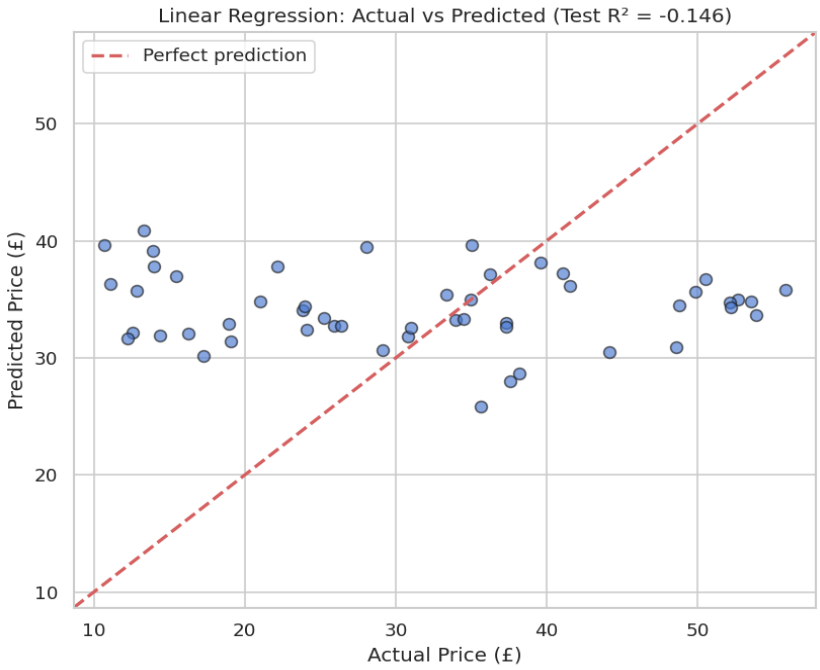


Figure 5 – Actual vs Predicted price on the held-out test set

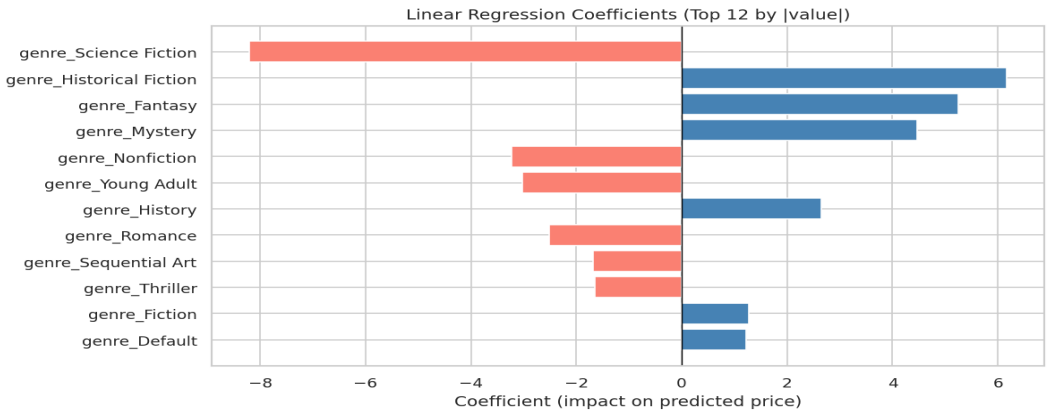


Figure 6 – Linear regression coefficients (top 12 by absolute magnitude)

Model limitations

- Synthetic nature of the website data limits real-world generalisability.
- Linear model cannot capture non-linear relationships or interactions beyond those implied by the encoding.
- Rare genres were filtered to avoid sparse dummy variables, discarding some category information.
- Important real-world drivers (author popularity, page count, publication year, competitor pricing) are absent.

6. Key Insights & Recommendations

Insights

- Price distribution is relatively flat; the sandbox does not reproduce the skewed price distributions typical of real book markets.
- Star rating and stock quantity exhibit only weak linear association with price – ratings remain useful for quality filtering but are poor proxies for expected price on this catalogue.
- Genre-level average prices differ modestly; sequential-art and certain non-fiction categories sit slightly above or below the overall mean.
- The transparent linear baseline is valuable precisely because it reveals the limited explanatory power of the currently available features.

Recommendations for stakeholders

Stakeholder	Recommendation
Online retailers	Treat rating and genre as secondary pricing signals; prioritise inventory turnover and promotional elasticity analysis.
Publishers / analytics teams	Enrich future models with author metrics, page count, publication date and competitor prices before production deployment.
Data-driven book buyers	Continue using star ratings for quality filtering, but do not infer price from rating alone.
Future modelling work	Expand the scrape to the full 1 000-book catalogue; experiment with regularised and tree-based regressors; engineer rating × genre interactions.

Table 4 – Stakeholder recommendations

7. Conclusion

This project delivered a complete, reproducible data-science pipeline from raw web data to actionable insights. Three hundred book records were scraped with BeautifulSoup, cleaned into a structured analytical dataset, explored through multiple visualisations, and modelled with linear regression. While the synthetic nature of the Books-to-Scrape catalogue constrains predictive power, the exercise demonstrates the full workflow expected of a junior data scientist and highlights the importance of richer feature engineering for real-world pricing applications.

Deliverables submitted: (1) Jupyter Notebook documenting the entire pipeline, (2) cleaned dataset `books_clean.csv`, (3) this professional project report.